

Kasi Ledger: Retail Management OS

Complete Functional Architecture, Operational Manual & Technical Specification

Platform: Android Native (Compose & Room SQLite) | Build: Production v2.4 | Date: September 2026

1. Executive Summary & Market Mission

Kasi Ledger is an enterprise-grade, offline-first point-of-sale, inventory intelligence, and alternative credit-scoring platform engineered specifically for independent retail merchants, spaza shop proprietors, general dealers, and kiosk operators in townships and informal commercial centers.

Historically, informal retail merchants transact predominantly in physical cash. Despite turning over substantial daily capital, they remain financially 'invisible' to commercial banks, micro-financiers, and working capital credit facilities. Kasi Ledger bridges this systemic divide by converting daily checkout scans, wholesale carton unbundling, and supplier invoice settlements into an auditable, digitally certified paper trail coupled with an objective 0 to 100 Alternative Credit Readiness Score.

Core Architectural Guarantees

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|-----------------------------------|--|
| - Offline-First Execution: | 100% self-contained on-device Room SQLite database. Zero cloud dependencies. |
| - Design System Palette: | Modern Ocean Blue (#2872A1) & Cloudy Sky (#CBDDE9) with high-contrast alert rose. |
| - Camera Vision Engine: | Embedded Google ML Kit barcode scanner supporting UPC, EAN-13, EAN-8, Code 128. |
| - Financial Scoring: | Rolling 30-day objective algorithm evaluating stability, margin health, and credit discipline. |
| - Certified PDF Exports: | Native Android A4 PDF generator with SHA-256 digital cryptographic verification hashes. |

2. Comprehensive Feature Breakdown by Module

Module A: Dashboard & Operational Cockpit

The Dashboard acts as the storekeeper's central morning cockpit, providing instantaneous store status, core financial KPIs, and fast shortcuts before trading starts:

- **Bespoke Hero Greeting & Status:** Displays storekeeper's personalized name, live store status indicator, and frosted geographic location badge.
- **Fast Operational Shortcuts:** One-tap direct access to 'Sell Items' (Cash Register) and 'Add Item' (Product Intake).
- **Today's Gross Turnover:** Aggregates total customer tender received during the current calendar day.
- **Today's Clean Net Profit:** Calculates true owner takeaway profit (Gross Sales minus Wholesale Inventory Cost).
- **Items Sold Counter:** Tracks physical piece volume moved across the counter in real time.
- **Stock Alerts Monitor:** Instant warning pill identifying items needing urgent wholesale replenishment.
- **Recent Transactions Log:** Interactive chronological feed of recent checkouts with item counts, payment method, and timestamps.

Module B: High-Throughput Point of Sale (POS) Register

Engineered for high-volume customer queues where rapid checkout is paramount to avoid queue congestion:

- **Multi-Modal Search & Filtering:** Real-time query bar matching product titles, barcodes, or custom SKUs, alongside horizontal category filter chips.
- **Rapid Touch Product Grid:** Visual cards showing high-resolution product thumbnails, live shelf stock count, unit price, and instant tap-to-cart.
- **Camera Barcode Scanning Modal:** Integrated camera overlay scanning standard retail barcodes with instant audio/haptic feedback.

- Cart Management Drawer: Line-item adjustments, bulk carton break warnings, item quantity steppers, and clear-cart safeguards.
- Tender Settlement Dialog: Supports Cash, Card, and Mobile Pay. Features one-tap quick-cash tender buttons (Exact, R50, R100, R200, R500) and automatic change calculation.
- Post-Sale Receipt & Sharing: Generates a formatted digital receipt with WhatsApp sharing and instant reset for the next customer.

Module C: Stock Control & Inventory Valuation

Provides total visibility into physical shelf stock and tied-up business working capital:

- Inventory Asset Valuation Header: Calculates total unique catalog SKUs, total physical shelf units, wholesale stock asset value (cost price), and potential retail revenue value.
- Low Stock Critical Filter: Dedicated quick-filter toggle isolating products at or below the minimum threshold.
- Inline Stock Steppers: Quick '+' and '-' buttons on product cards allowing cashier to adjust shelf stock immediately without opening edit forms.
- Profit Margin Badges: Color-coded margin percentage chips (Green for healthy profit $\geq 15\%$, Amber/Rose for slim or loss-leader items).
- Catalog Modification: Comprehensive edit dialog for re-pricing, barcode editing, and safe item removal.

Module D: Product Intake & Wholesale Crate Splitter

Addresses the common retail challenge of purchasing bulk cartons from wholesale depots and selling individual units:

- Visual Association: Attach custom photos via Android Photo Picker (zero-permission) or select from pre-configured FMCG grocery presets.
- Non-Barcoded Item Support: Toggle for loose fruit, bread, or single eggs that automatically generates a unique synthetic barcode identifier (e.g. NB-APPL-491).
- Dynamic Margin Simulator: Real-time profit calculator simulating exact profit per piece and profit margin percentage before saving.
- Wholesale Crate Splitter: Enter total box cost and pack size (e.g., R240 for 24 cans); automatically derives single-unit cost (R10.00) and updates stock count.

Module E: Financial Analytics & Business Intelligence

Visual analytics that transform raw receipts into strategic actionable business insights:

- Multi-Timeframe Scope: Toggle reports across Today, This Week, This Month, and All Time.
- Core Financial KPIs: Gross Turnover, Clean Net Profit, Cost of Goods Sold (COGS), Average Basket Size, and Overall Net Margin %.
- Interactive Sales Trend Chart: Custom vector canvas chart visualizing sales velocity with touch scrubber, smooth Bezier curves, and soft gradients.
- Departmental Category Share: Doughnut distribution chart and percentage table illustrating revenue drivers across product families.
- Stock Turnover Velocity Meter: Categorizes stock movement into Fast Turnover (High Liquidity), Optimal Turnover, and Stagnant Capital.

Module F: Alternative Credit Scoring & Banking Portal

The cornerstone innovation transforming unbanked informal trading into formal creditworthiness:

- Merchant KYC & GPS Anchoring: Captures business name, proprietor national ID/passport, phone number, physical address, and GPS coordinates.
- 30-Day Rolling Credit Scoring Engine: Scores merchants from 0 to 100 points based on three objective pillars: Turnover Stability (40 pts), Margin Health (30 pts), and Supplier Discipline (30 pts).
- Four-Tier Commercial Credit Tiers: Prime Merchant (85-100 pts), Bankable (70-84 pts), Emerging (50-69 pts), and Building (0-49 pts).
- Wholesaler Credit & Invoice Ledger: Tracks wholesale supplier invoices and builds repayment discipline as invoices are marked paid on time.

Module G: Native PDF Generation & Digital Certification

Native Android graphics document generator creating professional, print-ready A4 documentation:

- **Monthly Store Performance Report:** Detailed summary of monthly turnover, gross margin, wholesale costs, top revenue products, and category shares.
- **Certified Bank Credit Readiness Statement:** Formal multi-section document designed for loan officers, featuring Merchant KYC, Credit Scorecard, Cash Flow capacity, Inventory Collateral Appraisal, and Wholesaler Ledger.
- **SHA-256 Digital Verification Hash:** Embeds a cryptographic audit hash preventing unauthorized modification or tampering of banking statements.
- **Direct Sharing Integration:** One-tap export to WhatsApp, Email, Google Drive, or Bluetooth thermal POS printers.

3. Technical Architecture & Relational Data Model

Database Relational Entities (Room SQLite)	
- ProductEntity:	id, name, barcode, category, price, costPrice, quantity, isBulk, bulkPackSize, isNonBarcoded, image
- SaleEntity:	id, timestamp, totalAmount, paymentMethod, itemsJson (snapshot of items, qty, cost, price)
- StoreProfileEntity:	id, storeName, ownerName, nationalId, phone, location, gpsCoordinates, currencySymbol, creditScore
- WholesalerInvoiceEntity:	id, supplierName, invoiceNumber, amount, issueDate, dueDate, isPaid, paidDate, wasPaidOnTime

4. Security, Privacy & Offline Resilience

- **Zero Cloud Leakage:** All financial transactions, customer basket items, and inventory valuations remain strictly stored on-device.
- **No Cloud Dependency:** Functions seamlessly in rural areas, underground markets, and during prolonged load-shedding / power cuts.
- **Tamper-Evident Verification:** Statements contain an immutable SHA-256 signature calculated from the merchant's financial records.

5. Recommended Operational Workflows

- **Morning Opening:** Check Dashboard stock alerts to identify products requiring restocking from wholesale suppliers.
- **Stock Delivery Intake:** When unloading cartons from wholesalers, use the Crate Splitter to ensure unit cost is accurately logged.
- **Counter Sales:** Scan barcodes or use quick-touch cards; enter cash tendered to calculate exact customer change.
- **Monthly Bank Loan Application:** At month-end, generate the Certified Bank Credit Statement to provide proof of turnover when applying for working capital loans.